**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 201 Major Exam I Term 253

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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**1.** The length of the parametric curve

$$x = \sin^{-1} t, \quad y = \ln \sqrt{1 - t^2}, \quad 0 \le t \le \frac{1}{4}$$

is equal to

(a) $\frac{1}{2} \ln 2$

(b) $\ln\left(\frac{5}{4}\right)$

(c) $\frac{1}{2} \ln\left(\frac{4}{3}\right)$

(d) $\frac{1}{2} \ln\left(\frac{5}{3}\right)$

(e) $2 \ln\left(\frac{5}{3}\right)$

**2.** If $\langle a, b, c\rangle$ is the vector of magnitude $\frac{3}{4}$ in the opposite direction of the vector $\langle 2, -2, 1\rangle$, then $a + b + c =$

(a) $-\frac{3}{4}$

(b) $\frac{1}{2}$

(c) $1$

(d) $-\frac{1}{4}$

(e) $\frac{5}{4}$

**3.** The parametric curve

$$x = 4 + 2\cos t, \quad y = -1 + \sin t$$

represents

(a) a circle with center $(4, -1)$

(b) a parabola

(c) an ellipse

(d) a hyperbola

(e) a line segment

**4.** If the point $R\left(\frac{12}{5}, 2, \frac{11}{5}\right)$ lies three-fifths of the way from the point $P(0, 5, 1)$ to the point $Q(a, b, c)$, then $ac + b =$

(a) 12

(b) $\frac{1}{25}$

(c) 15

(d) 0

(e) $-4$

**5.** If the area of the triangle with the vertices $A(1, 2, 0)$, $B(0, 0, x)$, and $C(0, -1, 0)$ is 3, then $2x^2 =$

(a) 11

(b) 7

(c) 3

(d) 5

(e) 9

**6.** The slope of the tangent line to the polar curve $r = \sin\theta$ at the point corresponding to $\theta = \frac{\pi}{3}$ is equal to

(a) $-\sqrt{3}$

(b) $-\frac{\sqrt{3}}{2}$

(c) $-2\sqrt{3}$

(d) $2\sqrt{3}$

(e) $\frac{\sqrt{3}}{3}$

**7.** The area of the region inside the circle $r = 2\cos\theta$ and outside both circles $r = 2\sin\theta$ and $r = 1$ is equal to

(a) $\pi + 4$

(b) $\dfrac{3\pi}{2} - 1$

(c) $\dfrac{\pi}{2} + 1$

(d) $2\pi - \dfrac{1}{2}$

(e) $\dfrac{\pi}{4} + 1$

**8.** The projection of $\vec{u} = \langle 4, 2, 0\rangle$ onto $\vec{v} = \langle 2, 1, -1\rangle$ is equal to

(a) $\left\langle \dfrac{2}{3}, -\dfrac{5}{3}, \dfrac{4}{3} \right\rangle$

(b) $\left\langle \dfrac{5}{3}, \dfrac{5}{3}, -\dfrac{5}{3} \right\rangle$

(c) $\left\langle 1, \dfrac{1}{2}, -\dfrac{1}{2} \right\rangle$

(d) $\langle 2, 1, 0\rangle$

(e) $\left\langle \dfrac{10}{3}, \dfrac{5}{3}, -\dfrac{5}{3} \right\rangle$

**9.** Which one of the following sets is a parametrization of the curve $y = 2x - 3$, $x \geq 0$.

(a) $x = t,\ y = 2t - 3,\ t \geq 1$

(b) $x = t + 1,\ y = 2t - 1,\ t \in (-\infty, \infty)$

(c) $x = \dfrac{t+3}{2},\ y = t,\ t \geq 0$

(d) $x = \sin t,\ y = 2\sin t - 3,\ t \in (-\infty, \infty)$

(e) $x = t + 1,\ y = 2t - 1,\ t \geq -1$

**10.** The area of the region in the common interior of the polar curves $r = 1 + \cos\theta$ and $r = 1 - \cos\theta$ is equal to

(a) $\dfrac{3\pi}{2} - 2$

(b) $\dfrac{3\pi}{2} - 4$

(c) $\dfrac{\pi}{2} + 2$

(d) $3\pi - 4$

(e) $2\pi + 4$

**11.** The polar equation $r = \sec\theta - 2\cos\theta$, $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$, can be converted to the rectangular equation

(a) $y^2 = \dfrac{x^2(1-x)}{1+x}$

(b) $y^2 = x^2 - x^3$

(c) $y^2 = x^2$

(d) $y^2 = \dfrac{x(1+x^2)}{1-x}$

(e) $y^2 = \dfrac{x^2(1+x)}{1-x}$

**12.** The slope of the tangent line to the parametric curve

$$x = t - \sin t, \quad y = 1 - \cos t$$

at the point corresponding to $t = \dfrac{\pi}{2}$ is equal to

(a) 0

(b) 1

(c) $\dfrac{1}{2}$

(d) 2

(e) $-1$

**13.** An equation for the sphere having the points $(2, 1, 3)$ and $(1, 3, -1)$ as endpoints of a diameter is given by

(a) $4x^2 + 4y^2 + 4z^2 + 12x - 16y - 4z - 8 = 0$

(b) $4x^2 + 4y^2 + 4z^2 + 12x + 16y + 8z - 8 = 0$

(c) $4x^2 + 4y^2 + 4z^2 - 12x - 16y - 8z + 8 = 0$

(d) $4x^2 + 4y^2 + 4z^2 - 6x + 16y + 8z - 16 = 0$

(e) $4x^2 + 4y^2 + 4z^2 - 6x - 8y - 4z + 8 = 0$

**14.** A vector parallel to the tangent line to the graph of $f(x) = x^2\cos x$ at the point $(\pi, -\pi^2)$ is

(a) $\langle 2, 2\pi\rangle$

(b) $\langle \pi, -\pi^2\rangle$

(c) $\langle 1, 2\pi\rangle$

(d) $\langle -\pi, 2\pi\rangle$

(e) $\langle -1, 2\pi\rangle$

**15.** A vector that is orthogonal to both vectors $\vec{u} = \langle 2, -1, 3 \rangle$ and $\vec{v} = \langle 4, 5, 6 \rangle$ is

(a) $\langle 3, 0, -2 \rangle$

(b) $\langle -7, 0, 2 \rangle$

(c) $\langle 21, 0, 8 \rangle$

(d) $\langle -21, 24, 14 \rangle$

(e) $\langle -3, 1, 2 \rangle$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(d)** | Integrand $\frac{1}{1-t^2}$; $L = \frac{1}{2}\ln\frac{1+t}{1-t}\big\|_0^{1/4} = \frac{1}{2}\ln\frac{5}{3}$. |
| 2 | **(d)** | $\langle a,b,c\rangle = -\frac{1}{4}\langle 2,-2,1\rangle$, so $a+b+c = -\frac{1}{4}$. |
| 3 | **(c)** | $\frac{(x-4)^2}{4} + (y+1)^2 = 1$: an ellipse. |
| 4 | **(a)** | $\overrightarrow{PR} = \frac{3}{5}\overrightarrow{PQ}$: $Q = (4,0,3)$, so $ac+b = 12$. |
| 5 | **(b)** | $\frac{1}{2}\sqrt{10x^2+1} = 3 \Rightarrow 10x^2 = 35 \Rightarrow 2x^2 = 7$. |
| 6 | **(a)** | $\frac{dy}{dx} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$, at $\frac{\pi}{3}$ equals $-\sqrt{3}$. |
| 7 | **(e)** | $A_1 = 1 - \frac{\sqrt{3}}{2}$, $A_2 = \frac{\pi}{4} + \frac{\sqrt{3}}{2}$; total $\frac{\pi}{4} + 1$. |
| 8 | **(e)** | $\frac{10}{6}\langle 2,1,-1\rangle = \langle\frac{10}{3},\frac{5}{3},-\frac{5}{3}\rangle$. |
| 9 | **(e)** | All satisfy $y = 2x-3$; only $x = t+1$, $t \geq -1$ gives exactly $x \geq 0$. |
| 10 | **(b)** | $A = 4\int_0^{\pi/2}\frac{1}{2}(1-\cos\theta)^2 d\theta = \frac{3\pi}{2} - 4$. |
| 11 | **(e)** | $x^2(1+x) = y^2(1-x)$, so $y^2 = \frac{x^2(1+x)}{1-x}$. |
| 12 | **(b)** | $\frac{dy}{dx} = \frac{\sin t}{1-\cos t}$, at $t = \frac{\pi}{2}$ equals 1. |
| 13 | **(c)** | Center $(\frac{3}{2},2,1)$, $r^2 = \frac{21}{4}$; expands to $4x^2+4y^2+4z^2-12x-16y-8z+8 = 0$. |
| 14 | **(e)** | $f'(\pi) = -2\pi$, so $\langle 1,-2\pi\rangle \parallel \langle -1,2\pi\rangle$. |
| 15 | **(a)** | $\vec{u}\times\vec{v} = \langle -21,0,14\rangle = -7\langle 3,0,-2\rangle$. |